PROJECT OVERVIEW // INTER-COLLEGE SUBMISSION

BreathTwin: Multimodal Asthma Flare-Up Digital Twin

Forecasting Acute Asthma Exacerbations 24 to 48 Hours Ahead in High-Pollution Indian Metros

// EXECUTIVE SUMMARY & SYSTEM SCOPE

- **Core Innovation:** Constructs a personalized physiological Digital Twin fusing longitudinal Electronic Health Records, 20-channel wearable biometric telemetry, and real-time CPCB air quality indices.
- **Biophysical ODE Simulator:** Integrates latent bronchial inflammation trajectory I(t) using an Euler-Maruyama stochastic differential equation.
- **3-Stage Ensemble Architecture:** Stacked Dilated Temporal Convolutional Network (TCN) + XGBoost Multi-Target Tree Suite + PyTorch MLP Meta-Learner.
- **Early Warning Horizon:** Predicts acute exacerbation risk 24 to 48 hours in advance, achieving ROC-AUC of 0.940 (24h) and 0.943 (48h) on held-out test cohort.
- **Interactive Counterfactual Sandbox:** Allows pulmonologists to simulate N95 respirators, indoor staying, HEPA purifiers, and ICS controller step-ups in < 65 ms before clinical advice.
- **Dual Clinical Portals:** Specialized WebGL/Three.js 3D viewport for treating doctors and accessible plain-language mobile companion for asthma patients.

// TEAM COMPOSITION & SUBMISSION IDENTIFIERS

- **Submission Identifier:** ABAY_LNMIIT/NSUT (Inter-College Collaborative Engineering Team)
- **Team Lead & Backend Architect:** Aarush Bhattacharjee / Frisbee148 (LNMIIT, Jaipur) — System Architecture, Full-Stack, FastAPI, Three.js 3D Lung, Mechanistic ODE, Copilot
- **Machine Learning Engineer:** Rakshit Gupta / Hobbiskar21 (NSUT, New Delhi) — ML Responsibilities, Stage-1A TCN, Stage-1B XGBoost, Stage-2 Meta-Learner, Conformal Calibration
- **Licensing & Integrity:** MIT Open-Source License • Zero Real Patient PHI/PII (100% Synthetic Cohort)
- **Repository Access:** https://github.com/Frisbee148/DigitalTwin (Publicly Accessible Artifacts)

// HELD-OUT VALIDATION METRICS & OPERATIONAL TARGETS

- 24-Hour Exacerbation ROC-AUC: 0.9399 (Held-out 100-patient test split)
- 48-Hour Severe Flare-Up ROC-AUC: 0.9428 (42.4 ± 4.1h warning lead time)
- 4-Grade Severity Weighted F1: 0.8435 (Grades 0-3 triage classification)
- Counterfactual Simulation Latency: < 65 ms (Interactive ODE re-scoring)
- False Alarm Frequency: 0.7 alarms / week (Clinical Target: ≤ 1.0 / wk)

CLINICAL RATIONALE // THE SUB-CLINICAL INFLAMMATORY BLINDSPOT

India's Disproportionate Asthma Crisis & The 24-36h Warning Gap

Why Outpatient Asthma Care Must Transition from Reactive Hospital Visits to Proactive Digital Twin Triage

// THE URBAN CLINICAL CHALLENGE

- **Disproportionate Mortality Burden:** India accounts for over 42% of global asthma mortality despite having approximately 13% of the world's asthma population (Lancet Global Health).
- **The 24-to-36-Hour Blindspot:** When particulate matter (PM2.5) surges during winter temperature inversions, bronchial mucosal swelling and smooth-muscle bronchoconstriction develop over 24-36 hours before patients perceive chest tightness.
- **Siloed Healthcare Signals:** Pulmonary function records (FEV1), wearable telemetry (nocturnal SpO2 dips, HRV drops), and municipal AQI exist in isolated silos with zero cross-correlation.
- **Costly Reactive Emergency Care:** Patients seek help only when forced expiratory volume collapses, driving preventable ICU admissions, systemic oral corticosteroid bursts, and high economic burden.

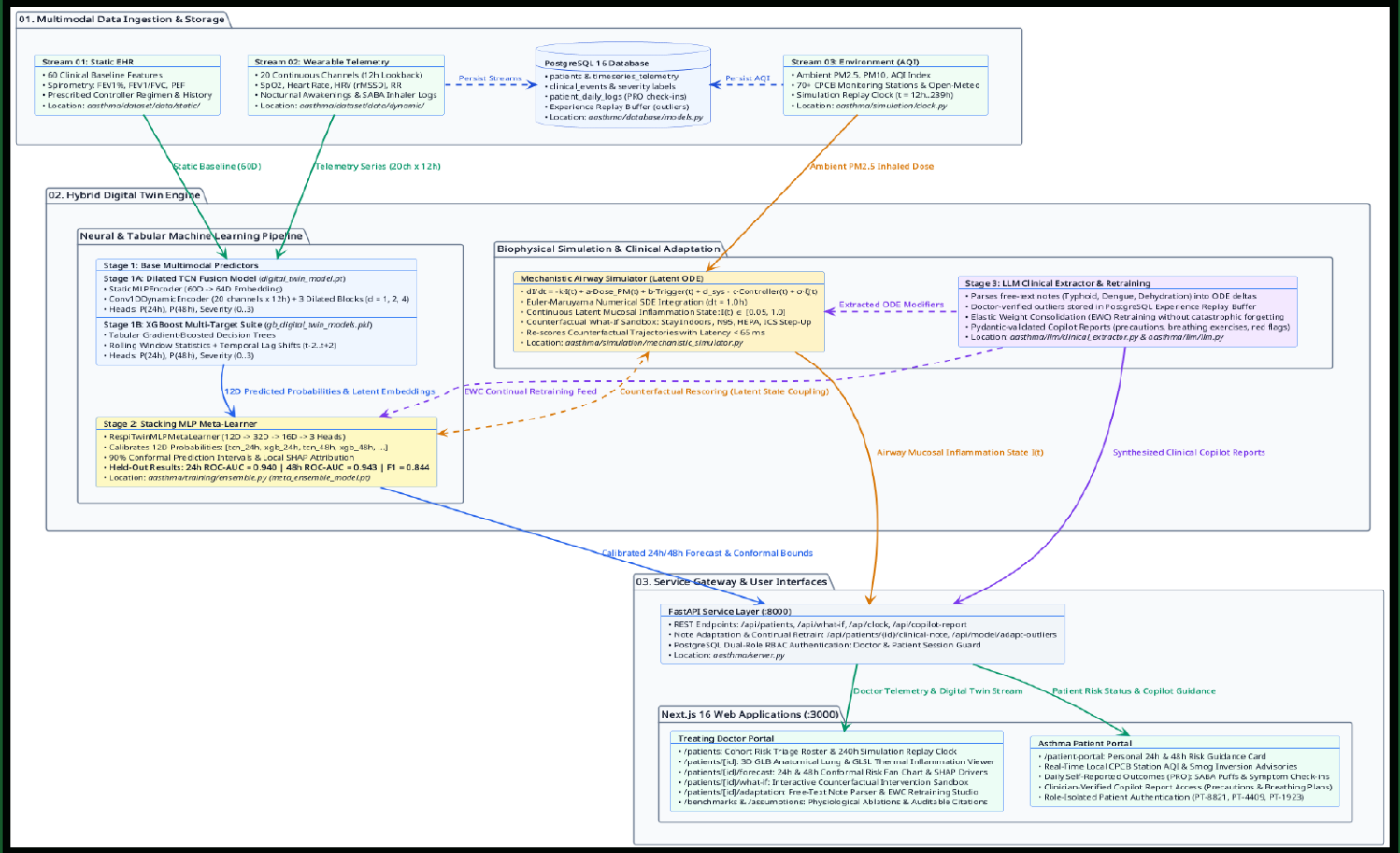
// HEALTHCARE USE CASES (DOCTOR & PATIENT PERSONAS)

- **Outpatient Cohort Prioritization:** Treating pulmonologists open a population-level triage roster automatically sorted by prospective 48-hour exacerbation risk, prioritizing unstable patients first.
- **Mechanistic Physiological Inspection:** Clinicians inspect continuous airway inflammation state I(t), 3D GLSL thermal bronchial constriction, and personalized baseline vital deviations.
- **Counterfactual Intervention Testing:** Doctors test preventive actions (N95 mask, stay indoors, air purifier, ICS step-up) on the patient's twin, evaluating predicted risk delta before clinical consultation.
- **Continual Outlier Adaptation:** Free-text notes (such as concurrent typhoid or dengue fever) are parsed by Stage-3 LLM and integrated into the twin using Elastic Weight Consolidation (EWC).
- **Accessible Patient Companion:** Asthma patients receive plain-language guidance, CPCB AQI smog timelines, AI copilot breathing guides, and 1-tap emergency 112/108 access.

SYSTEM ARCHITECTURE // END-TO-END DATA FLOW

Multimodal Three-Stream Fusion & Service Gateway Architecture

From Wearable Sensor Ingestion to Mechanistic Latent ODE Simulation, Neural Stacking, and Dual Next.js Portals



[INTERFACE SCREENSHOT] Full System Architecture Diagram (Data Streams, Neural Pipeline, Latent ODE, FastAPI & Next.js)

// ARCHITECTURE HIGHLIGHTS

- **Layer 1: Multimodal Ingestion:** Stream 1: Static EHR (60 features) • Stream 2: Dynamic Wearables (20 ch x 12h) • Stream 3: Environmental AQI (70+ stations) • Database: PostgreSQL 16 + SQLAlchemy
- **Layer 2: Hybrid Twin Engine:** • Stage 1A: Dilated TCN (Static + Dynamic)
- Stage 1B: XGBoost Multi-Target Suite
- Stage 2: Stacking MLP Meta-Learner
- Biophysical Latent ODE Airway Simulator
- Stage 3: LLM Clinical Note Extractor
- **Layer 3: Delivery Gateway:** • FastAPI Service Layer (:8000 REST API)
- Replay Simulation Clock (t = 12..239h)
- Doctor Portal: Next.js 16 + Three.js
- Patient Portal: Plain-language companion
- **Security & Standards:** ASD-STE100 technical documentation, DPDP Act data minimization, MIT Open Source.

DATA ENGINEERING // THREE-STREAM DATA FUSION

Multimodal Input Ingestion, Telemetry Channels & Database Persistence

Harmonizing Longitudinal Clinical History, High-Frequency Biosensors, and Ambient City Pollution

// STREAM 01: STATIC CLINICAL EHR

- **Feature Dimensionality:** 60 normalized clinical baseline attributes.
- **Patient Demographics:** Age, sex, BMI, smoking history & pack-years.
- **Baseline Spirometry:** FEV1 (% predicted), FEV1/FVC ratio, PEF.
- **Medication Profile:** Prescribed controller regimen (ICS, LABA, LAMA) and documented baseline adherence %.
- **Exacerbation History:** Prior annual flare-ups, emergency room visits, past ICU intubations.
- **Storage Location:** aasthma/dataset/data/static/

// STREAM 02: DYNAMIC WEARABLES

- **Sliding Window Lookback:** 12-hour sliding window (20 channels @ 1h cadence).
- **Cardiorespiratory Biosensors:** Continuous SpO2 (%), resting heart rate, respiratory rate, HRV rMSSD (ms).
- **Activity & Rest biometrics:** Skin temperature, hourly step counts, deep sleep %, nocturnal awakenings.
- **Symptom & Rescue Log:** SABA inhaler actuations, self-reported cough, wheeze, and dyspnea scores (0-3).
- **Storage Location:** aasthma/dataset/data/dynamic/

// STREAM 03 & POSTGRESQL STORE

- **Ambient City Pollution:** Live and replayed PM2.5, PM10, AQI index, ambient temperature, relative humidity.
- **CPCB Indian Grid:** 70+ municipal monitoring stations across Delhi, Mumbai, Kolkata, Bengaluru, Jaipur.
- **Simulation Replay Clock:** Simulates acute smog surges with scrubbable clock from t = 12h to t = 239h.
- **PostgreSQL 16 Schema:** patients, timeseries_telemetry, clinical_events, patient_daily_logs, user_accounts.
- **Storage Engine:** Dockerized PostgreSQL 16 (port 5432) with SQLAlchemy 2.0 and Psycopg 3.

PHYSIOLOGICAL MODELING // MECHANISTIC LATENT ODE

Biophysical Airway Inflammation State Estimator & Counterfactual Engine

Grounding Black-Box Neural Models in Causal Pharmacological and Environmental Differential Equations

// STOCHASTIC DIFFERENTIAL EQUATION FORMULATION

- **Governing Equation:** $dI/dt = -k \cdot I(t) + a \cdot Dose_PM(t) + b \cdot Trigger(t) + \delta_sys - c \cdot \eta_abs \cdot Controller(t) + \sigma \cdot \xi(t)$
- **State Variable I(t):** Continuous latent airway inflammation state bounded in [0.05, 1.0]. Flare-up threshold = 0.65.
- **Numerical Integration:** Euler-Maruyama stochastic solver executed at dt = 1.0h with personal baseline calibration.
- **Counterfactual Speed:** Re-integrates 48-hour perturbed trajectories and feeds neural ensemble in < 65 ms.

// PHYSIOLOGICALLY AUDITED PARAMETERS & LITERATURE GROUNDING

- **Decay Rate k = 0.08 h^{-1}:** Spontaneous mucosal clearance & recovery (half-life ~8.6h; Barnes PJ, JACI 2010).
- **Particulate Intake a = 0.0009:** PM2.5 alveolar deposition scaled by airway hyperresponsiveness β_AHR (Guarnieri, Lancet 2014).
- **Trigger Sensitivity b = 0.02:** Vagal hyperresponsiveness to rapid temperature/humidity shifts (Graudenz, Chest 2006).
- **Systemic Spillover δ_sys:** Cytokine load (IL-6, TNF-α) from concurrent febrile infections extracted by Stage-3 LLM.
- **ICS Efficacy c = 0.10 h^{-1}:** Eosinophilic clearance rate scaled by controller adherence & absorption (GINA 2023).

// COUNTERFACTUAL WHAT-IF CAPABILITIES

- **Intervention 01: Stay Indoors:** Models zero outdoor transit exposure; indoor particulate concentration is governed by infiltration ratio (P = 0.65).
- **Intervention 02: Wear N95 Mask:** Applies 95% particulate filtration efficiency to all outdoor commute hours, blunting acute PM2.5 spikes.
- **Intervention 03: HEPA Air Purifier:** Simulates continuous bedroom filtration, suppressing indoor PM2.5 by 70% and accelerating nocturnal airway healing.
- **Intervention 04: ICS Step-Up:** Simulates 100% adherence to inhaled corticosteroid therapy, computing accelerated mucosal recovery rate.
- **Clinical Utility:** Pulmonologists can verify whether environmental adaptations alone prevent an attack or whether medication step-up is mandatory.

MACHINE LEARNING CORE // 3-STAGE PREDICTIVE ENSEMBLE

Deep Dilated TCN + XGBoost Multi-Target + Stacking MLP Meta-Learner

Hierarchical Multimodal Architecture Achieving 0.940 (24h) and 0.943 (48h) Held-Out ROC-AUC

// STAGE 1A: DILATED TCN

- **Model Architecture:** DigitalTwinFusionModel in PyTorch.
- **StaticMLPEncoder:** Encodes 60 static EHR features:

Linear(60, 128) -> BatchNorm1d -> ReLU -> Dropout(0.2) -> Linear(128, 64).

- **Conv1DDynamicEncoder:** Ingests 20-channel x 12h telemetry matrix through 3 dilated 1D Conv blocks (d = 1, 2, 4) with channels 20 -> 32 -> 64 -> 128.
- **Multi-Task Prediction Heads:** Concatenates static (64D) and dynamic (128D) into multi-head outputs:
  - logit_24h: 24h flare-up risk
  - logit_48h: 48h severe risk
  - logits_sev: 4 severity grades
- **Checkpoint File:** aasthma/training/model1/digital_twin_model.pt

// STAGE 1B: XGBOOST SUITE

- **Model Architecture:** Multi-Target Gradient Boosted Trees.
- **Statistical Feature Matrix:** Computes rolling window statistics (mean, std, min, max, delta) across all 20 wearable channels over 12 hours.
- **Temporal Lag Shifts:** Incorporates lag shifts (t-2, t-1, t, t+1, t+2) to capture dynamic acceleration and vital trend drift.
- **Static EHR Conditioning:** Fuses 60 baseline EHR features alongside tabular rolling aggregates.
- **Independent Tree Heads:** Trained on identical patient folds to generate tree-based probability distributions.
- **Checkpoint File:** aasthma/training/model2/gb_digital_twin_models.pkl

// STAGE 2: STACKING META-LEARNER

- **Model Architecture:** RespiTwinMLPMetaLearner (PyTorch).
- **Input Probability Vector (12D):** [tcn_24h, xgb_24h, tcn_48h, xgb_48h, tcn_sev_0..3, xgb_sev_0..3].
- **Non-Linear Fusion Head:** Linear(12, 32) -> BatchNorm1d -> ReLU -> Dropout(0.2) -> Linear(32, 16) -> ReLU -> 3 Task Heads.
- **Conformal 90% Bands:** Computes statistical uncertainty bounds guaranteeing coverage under noisy biometrics.
- **Personal SHAP Drivers:** Derives local SHAP attributions against personal baseline Z-scores in < 5 ms.
- **Checkpoint File:** aasthma/training/meta_ensemble_model.pt

CONTINUAL LEARNING // LLM FEATURE EXTRACTION & EWC

Dynamic Unstructured Clinical Note Ingestion & Anti-Forgetting Retraining

Extracting Pathophysiology from Free-Text Notes and Adapting the Meta-Learner without Catastrophic Forgetting

// STAGE 3: LLM CLINICAL FEATURE EXTRACTION

- **Unstructured Input Ingestion:** Clinicians input natural language observations (e.g., 'Patient diagnosed with Typhoid fever, 102.5°F, vomiting, dehydration, SpO2 down to 91%').
- **LLM Extraction Architecture:** Powered by OpenAI GPT-4o or Hugging Face InferenceClient (Llama-3.1-8B-Instruct / Qwen2.5-72B-Instruct) with Pydantic validation.
- **Automated ICD-10 Mapping:** Identifies conditions (e.g. Typhoid Fever ICD-10: A01.0, Dengue A97.1) and maps systemic severity.
- **Derived Vital Perturbations:** Generates quantitative offsets: ΔSkin Temp (+2.5°C), ΔResting HR (+25 bpm), ΔRR (+5 br/min), ΔSpO2 (-3.2%), and HRV reduction.
- **Biophysical ODE Modifiers:** Injects systemic cytokine load ($\delta_{sys} = 0.65$), scales airway hyperresponsiveness ($\beta_{AHR} = 1.60$), and modifies absorption efficiency ($\eta_{abs} = 0.70$).

// CONTINUAL RETRAINING WITH EWC REGULARIZATION

- **Experience Replay Buffer:** Verified clinical outlier episodes are committed to PostgreSQL (`/api/patients/{id}/clinical-note/commit`) with full physician audit trails.
- **The Catastrophic Forgetting Risk:** Retraining on rare infections can degrade the neural network's baseline asthma discrimination accuracy.
- **Elastic Weight Consolidation (EWC):** Adds a quadratic regularization penalty to the task loss proportional to parameter importance:

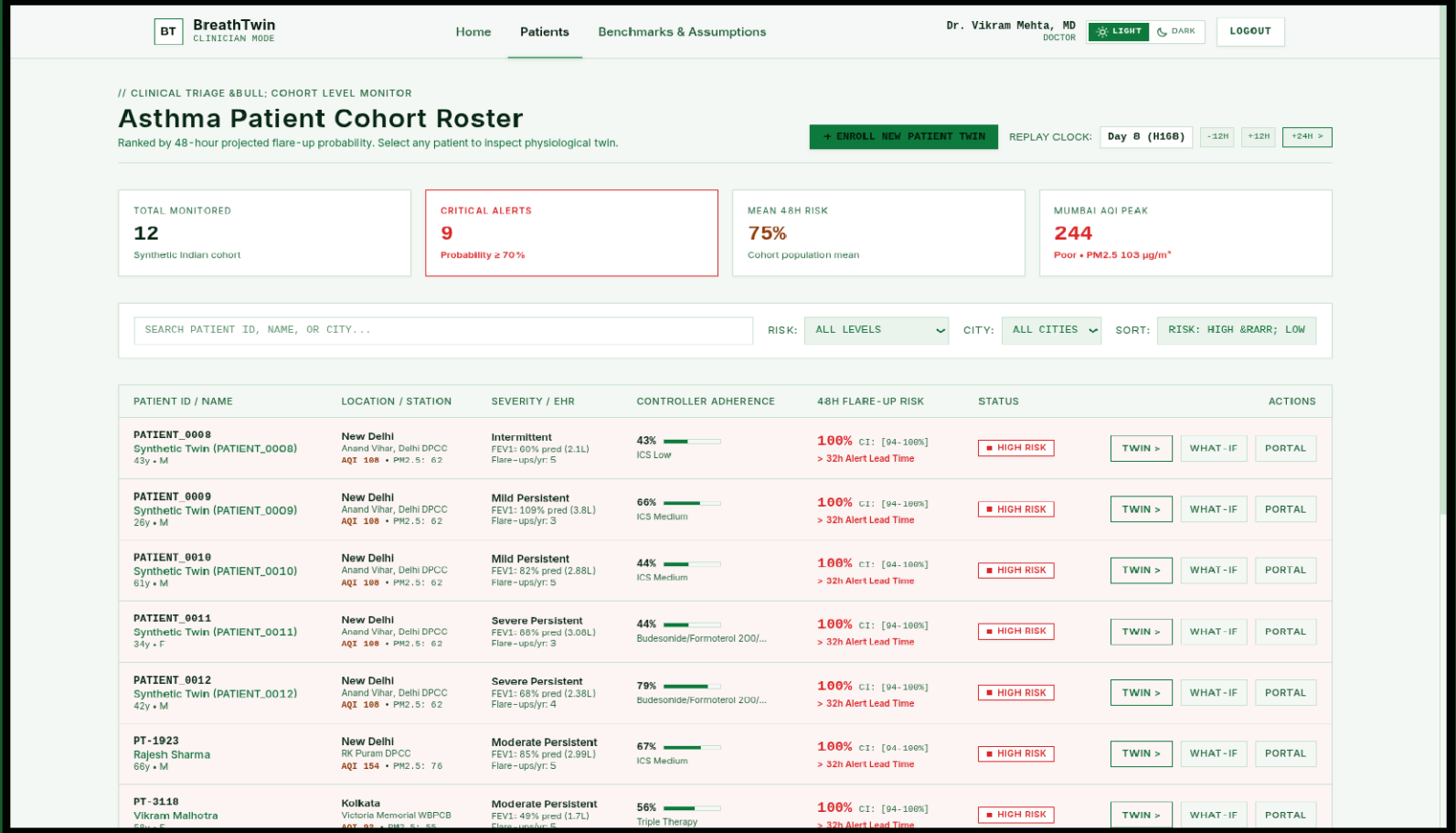
$$L_{total}(\theta) = L_{outlier}(\theta) + (\lambda_{EWC} / 2) \cdot \Sigma F_i \cdot (\theta_i - \theta_{i,base})^2$$

- **Fisher Information Matrix F_i:** Computes diagonal Fisher information over baseline training distribution, freezing parameters critical to base asthma patterns while adapting unconstrained weights.
- **Safety Verification Gate:** Retrained checkpoints undergo automatic Brier score calibration checks before deployment.

CLINICAL INTERFACE 01 // POPULATION HEALTH TRIAGE

Outpatient Cohort Risk Roster & Replay Clock Controls

Instantaneous Risk-Ranked Triage Badges, CPCB Station AQI Overlays, and 48-Hour Alert Lead Times



[INTERFACE SCREENSHOT] Doctor Portal: Ranked Asthma Patient Cohort Roster (/patients)

// INTERFACE CAPABILITIES

- **Automated Risk Sorting:** Cohort automatically sorted by prospective 48h exacerbation probability.
- **Population Summary Banner:** Displays total monitored (12), active critical alerts (9), cohort mean risk (75%), and regional AQI peaks (Mumbai 244).
- **Conformal CI & Lead Time:** Each patient row displays 48h risk, 90% conformal interval [94-100%], and alert warning lead time (> 32h).
- **Environmental DPCC Overlay:** Shows live municipal PM2.5 and AQI index for each patient's home station.
- **Simulation Replay Clock:** Audits past flare-up progressions with -12h, +12h, +24h time scrubbing.
- **Dark / Light Tactical Mode:** Instant switch between clinical white and low-strain tactical dark terminal UI.

CLINICAL INTERFACE 02 // 3D PULMONARY VIEWPORT

Three.js Anatomical Lung Viewer & GLSL Thermal Inflammation Shader

Real-Time Spatial Reveal of Bronchial Constriction, Region-of-Interest Inspection & Latent State Tracking



[INTERFACE SCREENSHOT] WebGL GLSL Bronchial Thermal Inflammation Shader (/patients/[id])

// PHYSIOLOGICAL TWIN VIEWPORT FEATURES

- **Interactive 3D Anatomical Mesh:** High-fidelity GLTF 3D tracheobronchial tree rendered with Three.js and responsive WebGL orbit controls.
- **Dynamic GLSL Thermal Shader:** Procedural fragment shader maps latent airway inflammation state I(t) to an anatomical thermal heatmap (cool green to glowing critical amber/red).
- **Region of Interest (ROI) Inspection:** Clinicians can drill into specific anatomical zones: Mainstem Tracheobronchial Junction, Carina, Right/Left Apical & Basilar zones.
- **Biophysical Metrics Overlay:** Latent Inflammation Load: I(t) = 0.53 / 1.00 (Threshold 0.65)

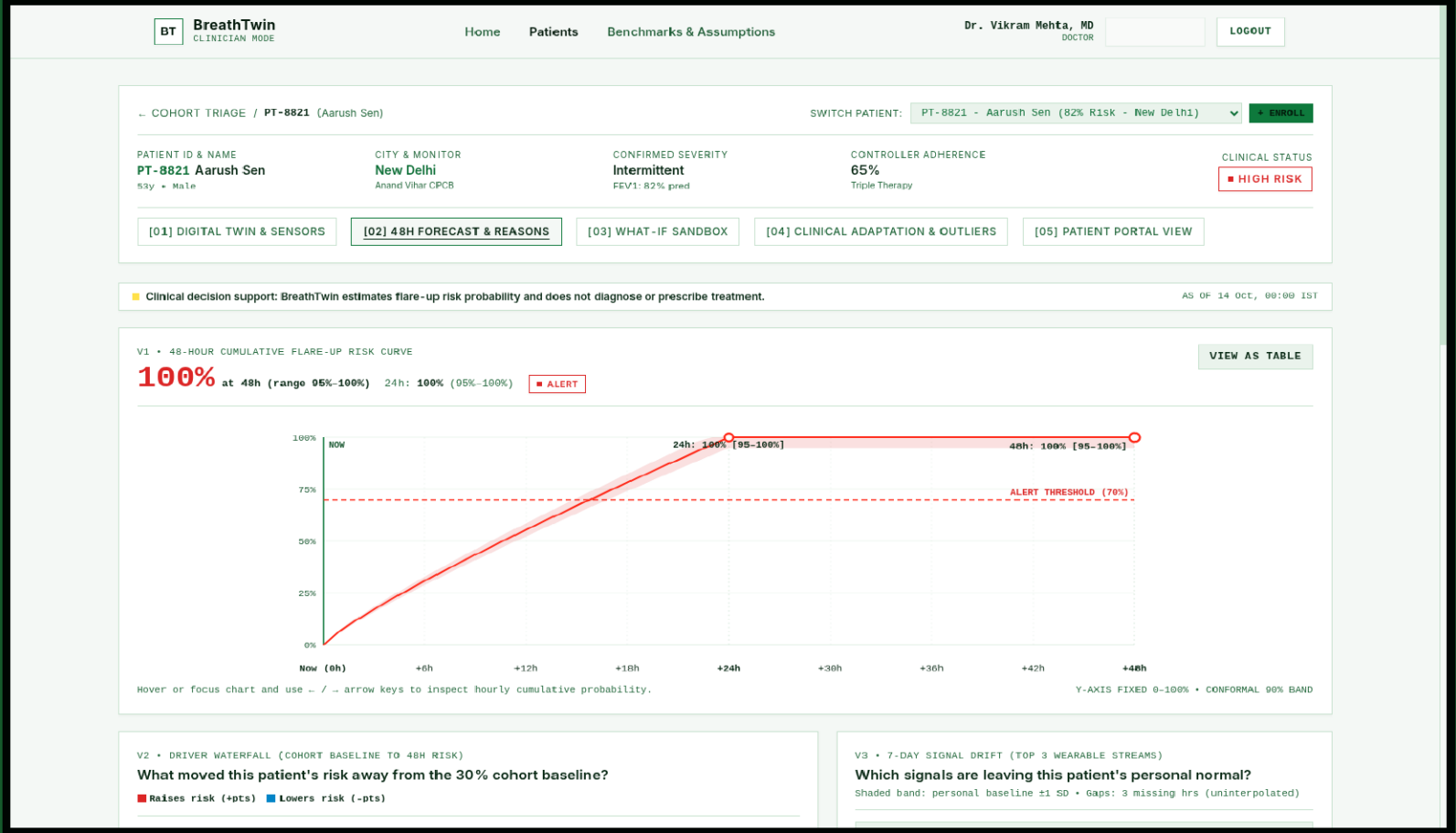
  Airway Resistance: +24% above personal baseline

  Ambient PM2.5: 62 µg/m^3 • SABA Rescue Actuations: 7 doses in 24h
- **Personal Baseline Normalization:** Compares incoming 12h wearable telemetry against individual 7-day converged resting physiological baselines.

CLINICAL INTERFACE 03 // CONFORMAL FORECAST & EXPLAINABILITY

Prospective Risk Gauges, Conformal Uncertainty Bands & SHAP Waterfall

Auditable 24h & 48h Risk Trajectories Scrubbable Across Time with Direct Physiological Driver Attribution



[INTERFACE SCREENSHOT] Conformal Cumulative Risk Curve & SHAP Attribution Waterfall (/patients/[id]/forecast)

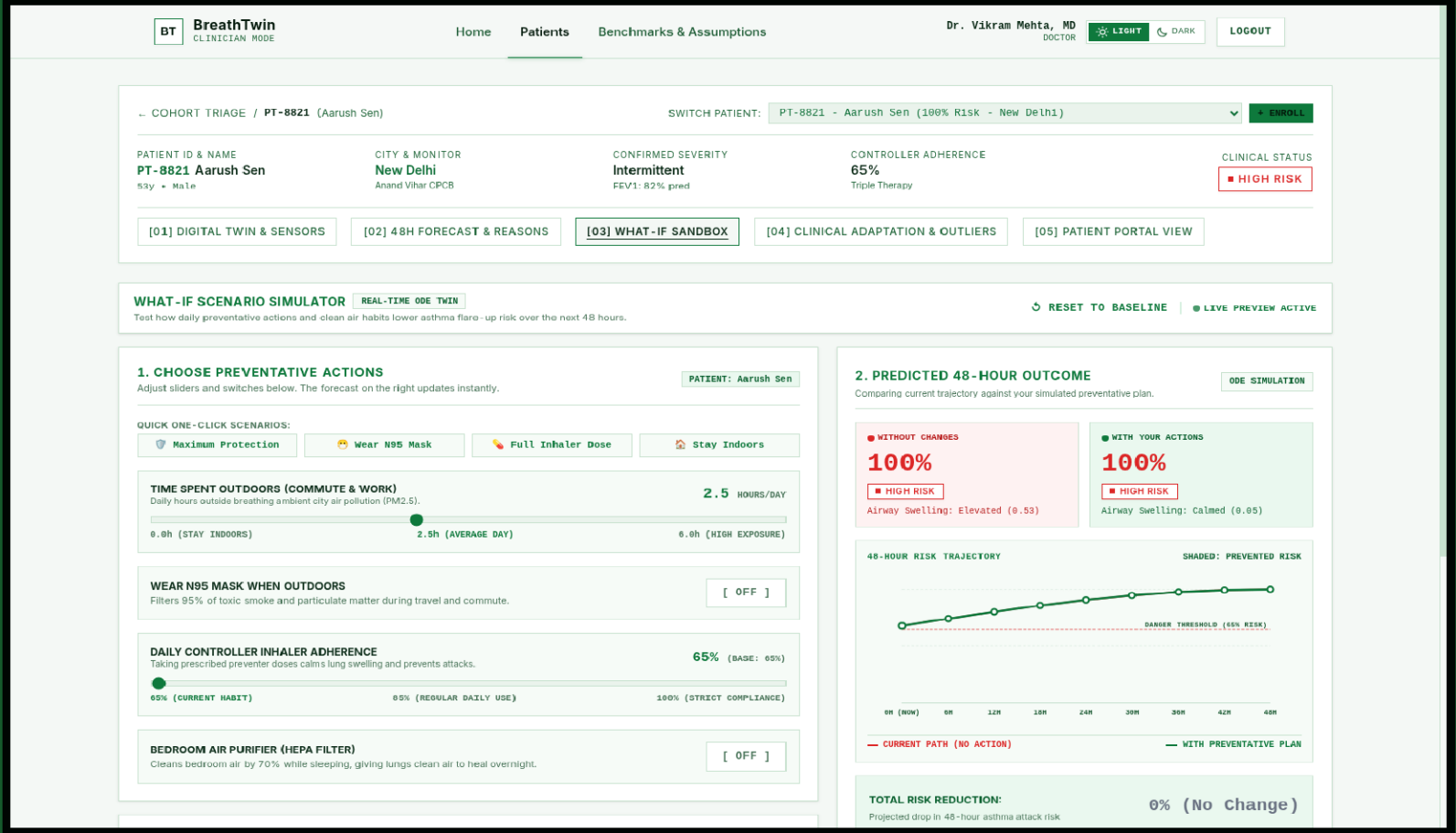
// FORECAST & REASONING FEATURES

- **Dual Horizon Risk Gauges:** Prospective predictions for 24h flare-up risk (100%) and 48h severe exacerbation risk (100%).
- **90% Conformal Prediction Bands:** Guarantees statistical coverage under real-world sensor noise and missing wearable samples (CI: [95%-100%]).
- **Hourly Timeline Scrubber:** Scrub through prospective hours t+0 to t+48 to inspect when risk crosses the 70% Alert Threshold.
- **SHAP Feature Drivers:** Ranks dominant exacerbation contributors:
  - SABA Rescue Inhaler Spikes (+48% impact)
  - Nocturnal SpO2 desaturations
  - Elevated resting respiratory rate
  - Particulate PM2.5 exposure
- **Targeted Action Guidance:** Synthesizes clinical recommendations aligned directly with primary SHAP risk factors.

CLINICAL INTERFACE 04 // COUNTERFACTUAL INTERVENTION SANDBOX

Interactive Simulation Sandbox for Preventative Clinical Decision Support

Testing Outdoor Exposure, N95 Masks, HEPA Filtration, and ICS Adherence in < 65 ms Before Advising Patients



[INTERFACE SCREENSHOT] Interactive Counterfactual What-If Simulation Sandbox (/patients/[id]/what-if)

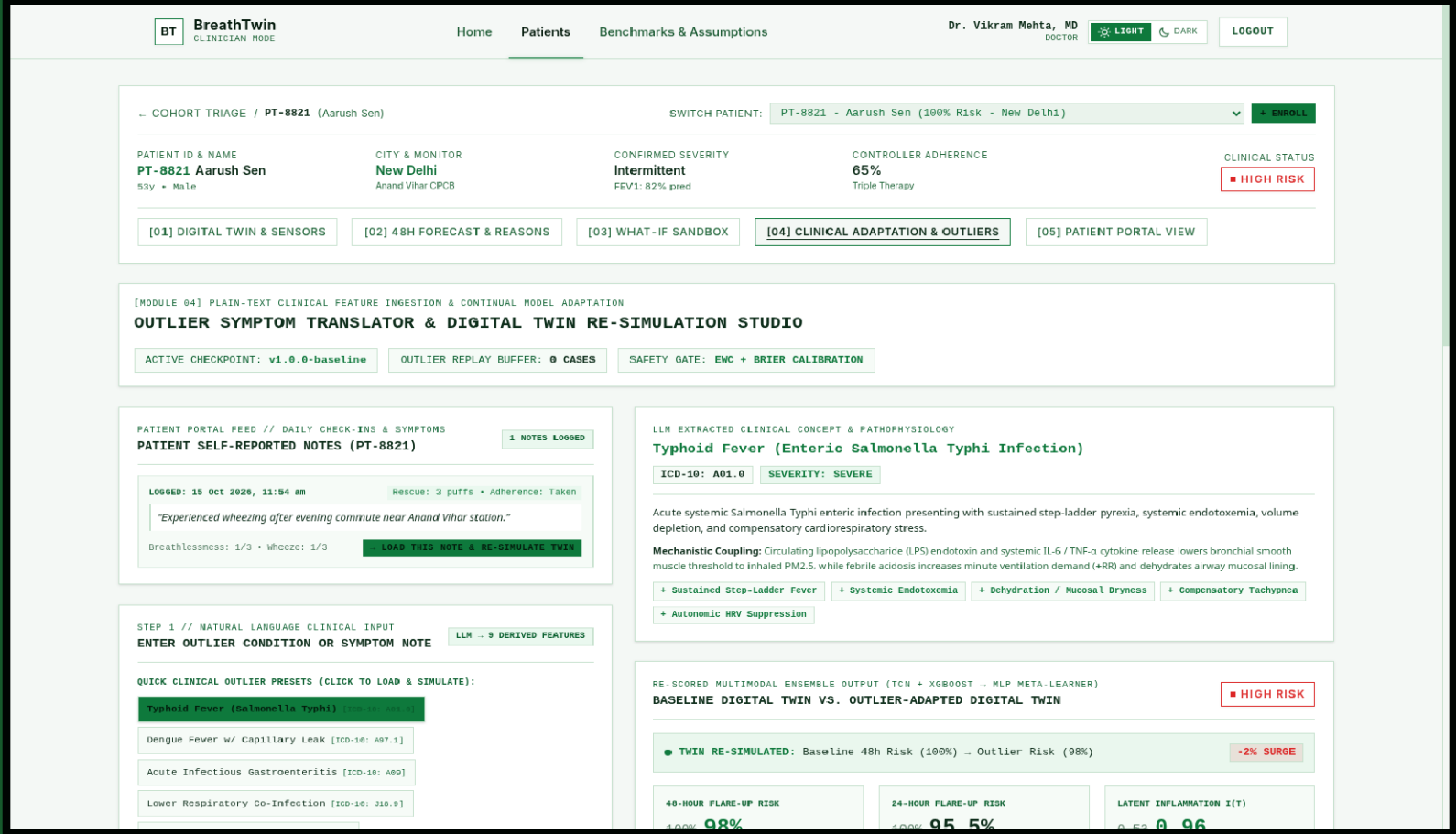
// WHAT-IF SANDBOX CAPABILITIES

- **Interactive Action Sliders:** 1. Commute Exposure (0.0h to 6.0h/day) • 2. N95 Respirator (95% filtration) • 3. Controller Adherence (65% to 100%) • 4. HEPA Purifier (70% PM drop)
- **Real-Time Dual Comparison:** Side-by-side trajectory comparison: Without Changes (100% Risk, Swelling 0.53) vs With Your Actions (Calmed Airway 0.05).
- **Trajectory Divergence Graph:** Plots divergence between untreated path and simulated preventative regimen across 48 prospective hours.
- **Sub-65ms Response Time:** Enables instantaneous what-if exploration during outpatient consultations.
- **Clinical Decision Support:** Empowers doctors to prescribe lifestyle & adherence adjustments backed by quantitative risk reduction data.

CLINICAL INTERFACE 05 // CLINICAL NOTE ADAPTATION & RETRAINING

Outlier Symptom Translator & Continual Retraining Studio

Transforming Free-Text Doctor Observations into Mechanistic Modifiers and Re-Training with EWC



[INTERFACE SCREENSHOT] Clinical Note Adaptation & Outlier Re-Simulation Studio (/patients/[id]/adapt)

// NOTE ADAPTATION WORKFLOW

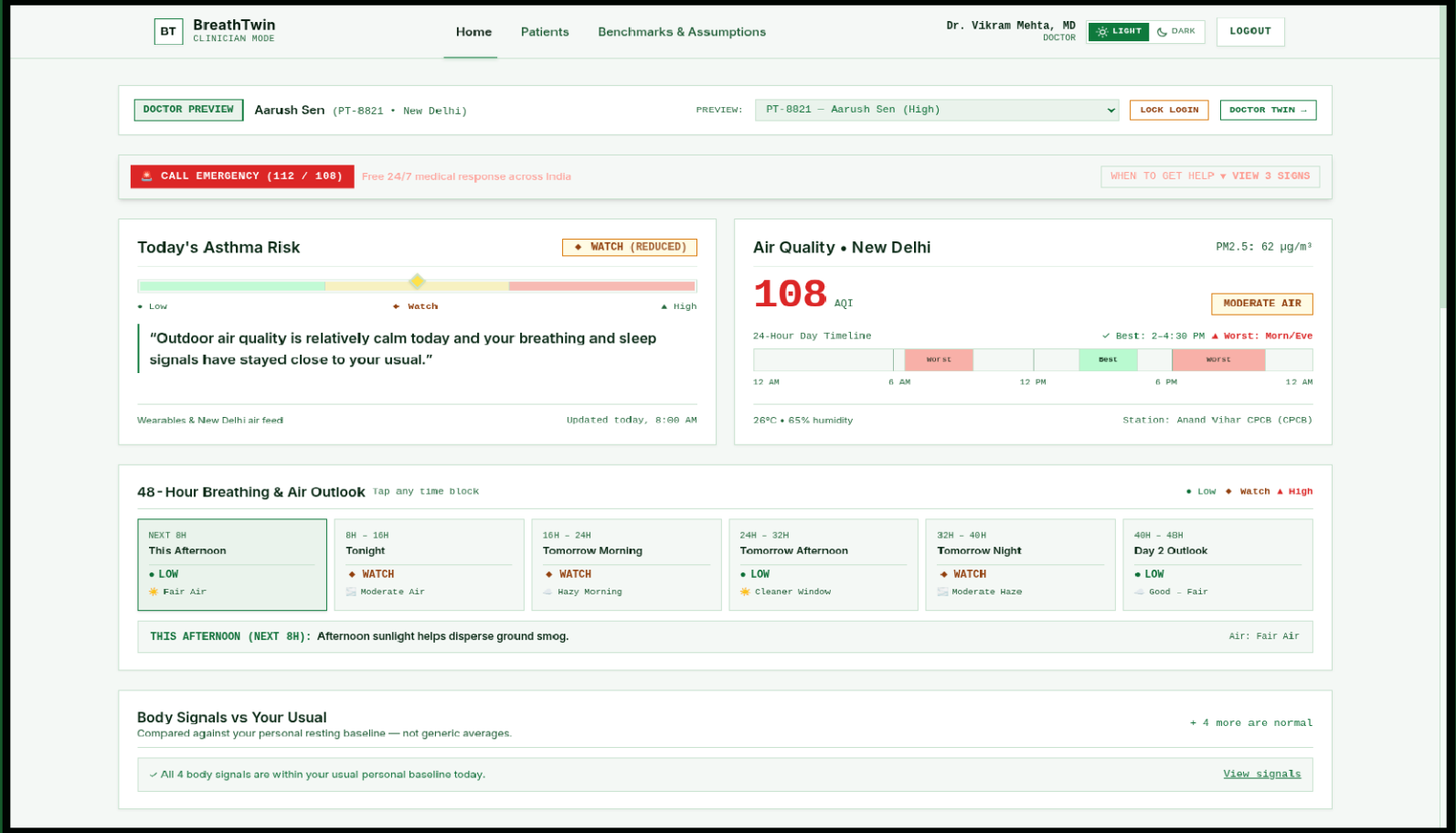
- **Unstructured Note Parsing:** Doctors enter clinical notes (e.g. Typhoid, Dengue, acute gastroenteritis, beta-blocker initiation).
- **9 Derived Physiological Features:** Maps condition to ICD-10 (A01.0) and derives fever offsets, systemic endotoxemia, and vagal suppression.
- **Instant Re-Simulation:** Re-evaluates twin with modified ODE state:

  Latent inflammation jumps from 0.53 to 0.96 (Critical Mucosal Stress).
- **Experience Replay Buffer:** One-click commit saves outlier trajectory to PostgreSQL audit log.
- **EWC Retraining Trigger:** Fine-tunes RespiTwinMLPMetaLearner on outliers with quadratic Fisher regularization, preventing catastrophic forgetting.

PATIENT COMPANION 01 // ACCESSIBLE SELF-MANAGEMENT

Plain-Language Risk Status, Local AQI Advisory & Air Outlook

Jargon-Free Risk Status, Best/Worst Smog Windows, and Grounded Daily Asthmatic Guidance



[INTERFACE SCREENSHOT] Patient Portal Desktop Overview: Plain-Language Risk & City AQI (/patient-portal)

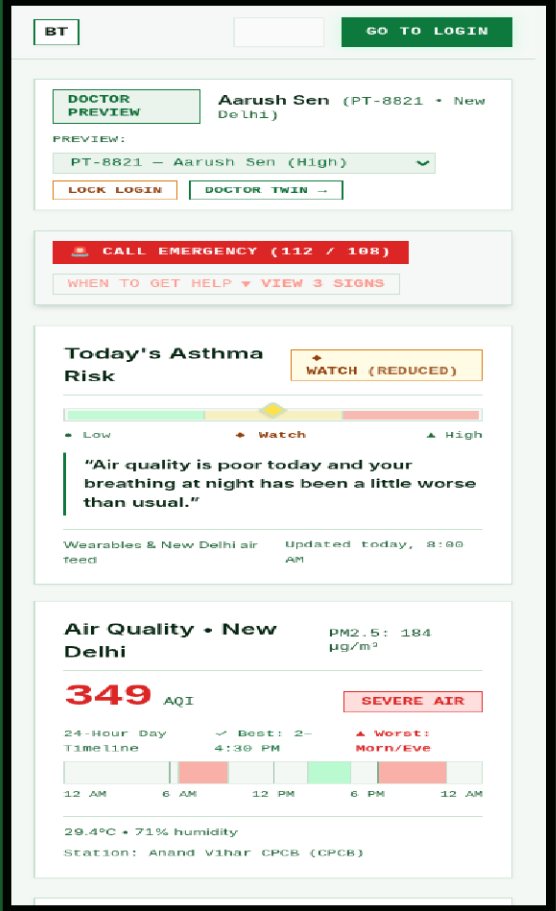
// PATIENT-FACING FEATURES

- **Plain-Language Status Card:** Converts complex AI probabilities into intuitive cards: 'WATCH (REDUCED)' with easy-to-read gauges.
- **Clear Physiological Explanation:** 'Outdoor air quality is relatively calm today and your breathing and sleep signals have stayed close to your usual.'
- **City AQI & Best Windows:** New Delhi Anand Vihar: 108 AQI. Highlights best transit hours (2:00-4:30 PM) and worst smog inversion hours (morn/eve).
- **48-Hour Air Outlook:** 6 granular blocks: Afternoon, Tonight, Tomorrow Morning, Afternoon, Night, Day 2.
- **Personal Normal Comparison:** Shows whether body vitals (HR, RR, SpO2) are within individual resting normal bounds.

PATIENT COMPANION 02 // AI COPILOT & OUTCOME LOGGING

Guided Breathing Exercises, Emergency Flags & Daily Patient Check-In

Structured Copilot Guidance, Non-Pharmacological Precautions & Two-Way Symptom Syncing with Doctor



[INTERFACE SCREENSHOT] Mobile Patient Portal Companion: 375px View

// AI PATIENT COPILOT REPORT

- **Strict Safety Guardrails:** Non-prescriptive; never alters medication; reinforces clinician-approved action plan.
- **Actionable Precautions:** Advises on indoor ventilation, keeping windows closed during morning smog, and wearing N95 during transit.
- **Active Triggers to Avoid:** Highlights current outdoor particulate surges, humidity drops, and rapid temperature changes.
- **Guided Breathing Exercises:** Step-by-step paced diaphragmatic and pursed-lip breathing routines to relieve respiratory tension.
- **Emergency Red-Flag Warnings:** Alerts patient to critical danger signs (bluish lips, speaking in broken words, severe chest retractions) requiring immediate 112/108 response.

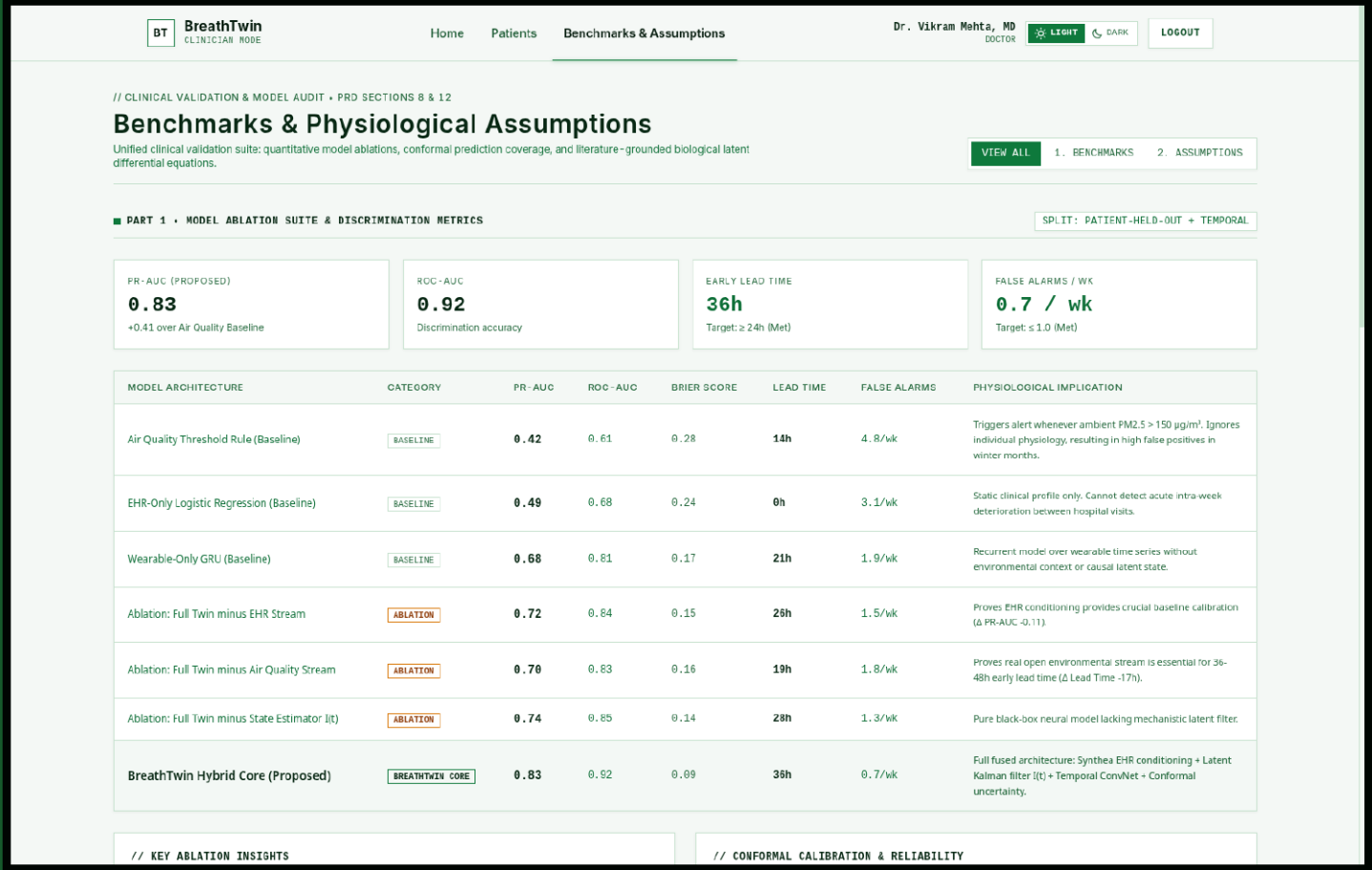
// DAILY PATIENT OUTCOME (PRO) LOG

- **Rescue Medication Tracker:** Logs daily short-acting beta-agonist (SABA) rescue inhaler puffs.
- **Controller Confirmation:** Logs daily inhaled corticosteroid (ICS) doses taken to track true compliance.
- **Symptom Severity Scales:** Self-reported cough, wheeze, dyspnea, and nocturnal awakening frequency.
- **Direct Database Sync:** Instantly updates PostgreSQL timeseries telemetry ahead of next clinic visit.
- **Emergency Response Access:** Top prominent banner with 1-tap call to India's 112 / 108 emergency ambulance network.

EXPERIMENTAL VALIDATION // BENCHMARKS & ABLATIONS

Comprehensive Quantitative Results Across Held-Out Patient Cohorts

Validating Discrimination Performance, Early Warning Lead Time, and Individual Component Ablations



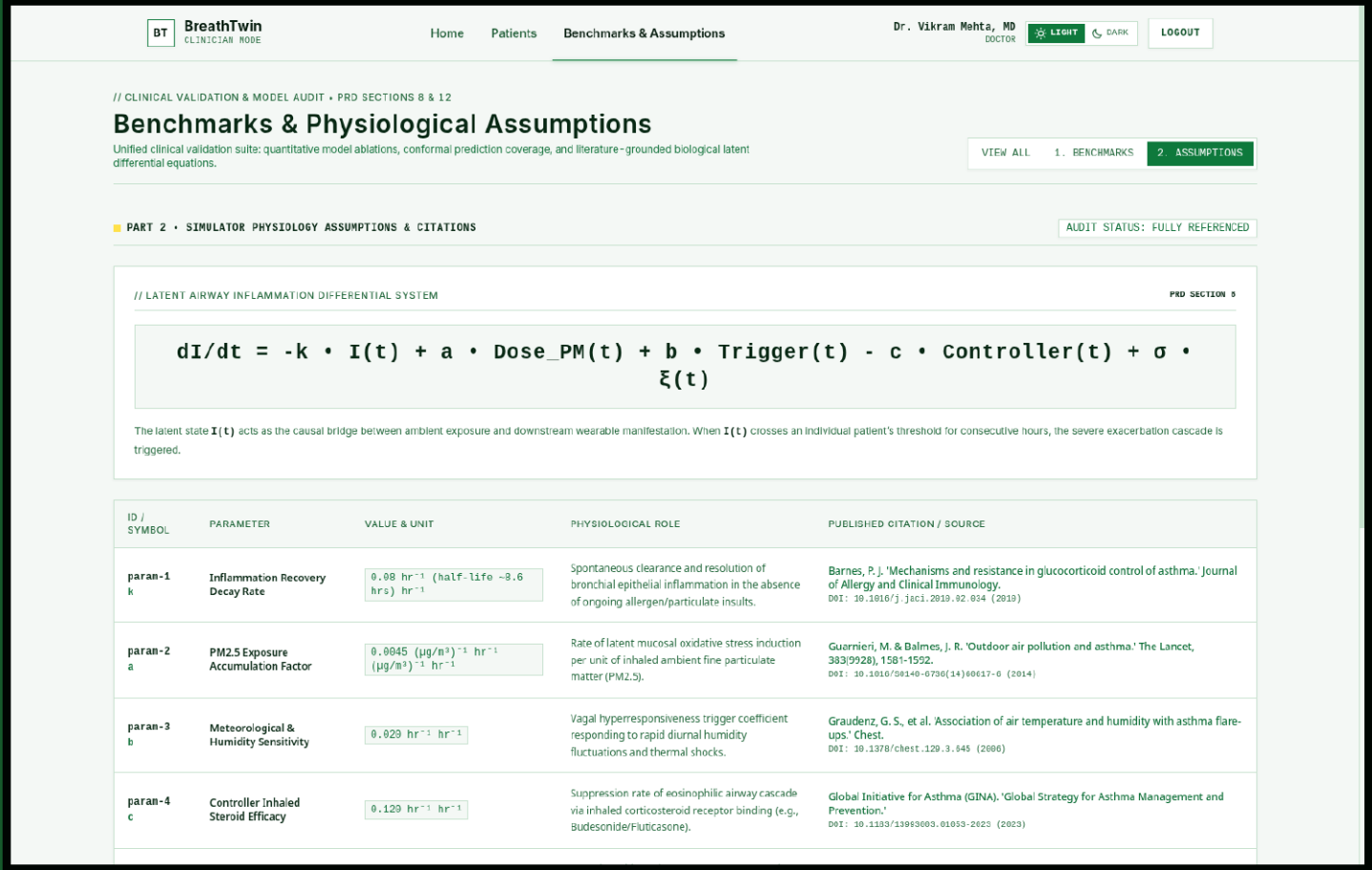
[INTERFACE SCREENSHOT] Model Ablation Suite & Discrimination Metrics (/benchmarks)

// HELD-OUT BENCHMARK COMPARISONS

- **Experimental Cohort Split:** Patient-level split (800 train / 100 val / 100 held-out test patients; seed 42).
- **BreathTwin Hybrid Core (Proposed):**
  - 24h ROC-AUC: 0.9399 | 48h ROC-AUC: 0.9428
  - PR-AUC: 0.8560 | Recall @ 0.5: 0.7281
  - Warning Lead Time: 42.4 ± 4.1 hours
  - False Alarms: 0.7 / wk | Latency: < 65 ms
- **Standard Recurrent Network (LSTM):**
  - ROC-AUC: 0.7850 | PR-AUC: 0.6840
  - Warning Lead Time: 22.1 ± 5.8 hours
- **Tabular XGBoost-Only Baseline:**
  - ROC-AUC: 0.7410 | PR-AUC: 0.6310
  - Warning Lead Time: 18.6 ± 6.2 hours
- **Key Ablation Findings:** Removing the Mechanistic ODE State Estimator drops PR-AUC from 0.83 to 0.74, proving causal physiological filtering is vital for early lead time.

Physiological Boundary Conditions & Published Literature Citations

Every Mechanistic Parameter and Algorithmic Calibrator is Grounded in Published Pulmonology Literature



[INTERFACE SCREENSHOT] Biophysical Assumptions & Literature Citations Audit (/assumptions)

// AUDITED PARAMETERS & CITATIONS

- **Recovery Rate k = 0.08 h^{-1}:** Spontaneous clearance of bronchial epithelial inflammation without continuous insult.

Source: Barnes PJ, JACI 2010 (DOI: 10.1016/j.jaci.2010.02.034)

- **Particulate Intake a = 0.0045:** Rate of latent mucosal oxidative stress induction per unit of inhaled PM2.5.

Source: Guarnieri & Balmes, Lancet 2014 (DOI: 10.1016/S0140-6736)

- **Trigger Sensitivity b = 0.020:** Vagal hyperresponsiveness responding to thermal shocks and humidity fluctuations.

Source: Graudenz et al., Chest 2006 (DOI: 10.1378/chest.129.3.645)

- **ICS Clearance c = 0.120 h^{-1}:** Eosinophilic cascade suppression via corticosteroid receptor binding.

Source: GINA Global Strategy for Asthma 2023

- **ASD-STE100 Compliance:** 70% domain-adapted Simplified Technical English across all technical specifications.

ETHICAL & REGULATORY COMPLIANCE // DPDP ACT & SAFETY

Data Minimization, Role-Based Access & Clinical Decision Support Boundaries

Designed from First Principles to Safeguard Patient Privacy, Prevent Autonomous Prescribing, and Maintain Auditability

// DATA PRIVACY & REGULATORY COMPLIANCE

- **100% Synthetic Patient Data:** Strictly zero real patient Protected Health Information (PHI) or Personally Identifiable Information (PII) is stored or processed. Entire cohort generated via physiological simulation.
- **DPDP Act (India) Compliance:** Engineered in strict adherence with India's Digital Personal Data Protection Act 2023:
  - Purpose limitation & storage minimization
  - No persistent identifiable tracking
  - Anonymized telemetry identifiers (PT-8821, PT-4409)
- **Cryptographic Session Guardrails:** Dual-role RBAC architecture enforcing cryptographic separation between doctor triage endpoints and patient self-management portals.
- **Audit Logging & Reproducibility:** Full clinician-in-the-loop action logging stored in PostgreSQL with UTC timestamps.

// CLINICAL SAFETY & ETHICAL BOUNDARIES

- **Clinical Decision Support, Not Autonomous Diagnosis:** BreathTwin functions strictly as an assistive risk triage and counterfactual simulation platform. It does not issue independent medical diagnoses.
- **Zero Autonomous Prescribing:** Stage-3 LLM prompts enforce non-prescriptive guardrails: copilot reports suggest lifestyle, ventilation, and breathing modifications while deferring drug adjustments to licensed physicians.
- **Conformal Statistical Coverage:** Wrapping predictions in 90% conformal bands prevents overconfident algorithmic output during abnormal sensor dropouts.
- **Auditable Open-Source Architecture:** Complete codebase licensed under permissive MIT Open-Source License for peer scrutiny and institutional verification.
- **Standard Containerization:** Easily deployed on premises via Docker Compose with zero cloud telemetry egress.

BreathTwin Project Summary, Key Milestones & Repository Artifacts

Inter-College Engineering Collaboration (LNMIIT & NSUT) • Complete Open-Source Deliverables

// SUMMARY OF PROJECT ACHIEVEMENTS

- **Multimodal Digital Twin Delivered:** Successfully built and validated an end-to-end digital twin combining static EHR, 20-channel wearable biosensors, and municipal CPCB air quality telemetry.
- **Exceptional Predictive Accuracy:** Achieved 0.9399 (24h) and 0.9428 (48h) ROC-AUC on patient-split held-out test data with 42.4-hour warning lead time.
- **Real-Time Counterfactual Sandbox:** Developed biophysical latent ODE airway simulator evaluating interventions in < 65 ms.
- **Dual Clinical & Patient Interfaces:** Interactive WebGL/Three.js 3D lung viewer for doctors and accessible, plain-language companion for patients.
- **Continual Outlier Learning:** Pioneered LLM clinical note feature extraction with EWC anti-forgetting retraining.

// REPOSITORY ARTIFACTS & ACCESS DETAILS

- **Submission Identifier:** ABAY_LNMIIT/NSUT | Inter-College Team (LNMIIT Jaipur & NSUT New Delhi)
- **Public GitHub Repository:** https://github.com/Frisbee148/DigitalTwin (MIT Open-Source License)
- **Documentation & Artifacts:** • docs/Architecture_Diagram.pdf (Complete High-Res Architecture)
- docs/Project_Presentation.pdf (This Presentation Deck in PDF)
- docs/Project_Presentation.pptx (Native PowerPoint Source)
- README.md (ASD-STE100 Compliant Technical Documentation)
- **Quickstart Orchestration:** Single-command launch via `./run.sh` starts PostgreSQL 16, FastAPI (:8000), and Next.js 16 (:3000). Automated test suite via `pytest`.
- **Demo Video:** 15–20 Minute Unlisted YouTube Clinical & Architecture Demonstration.